



SUPERIOR UNIVERSITY

Mobile Application Development

Final Milestone Report

Submitted To: Miss Esha

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FINAL MILESTONE

1. Introduction

Lab 13 focuses on preparing a completed mobile application for production release. The main objective is to understand the deployment workflow, configure the application for release, generate a signed build, test it on a real device, and prepare the required publishing material. In this lab, the Android publishing path was followed because the application was developed using Flutter and tested on a physical Android device.

The selected application is Persist, a behavior-focused goal tracking mobile app. Persist helps users create manual or AI-generated goals, complete daily tasks, log mood after task completion, view behavior insights, estimate skip risk, and use an AI reflection coach for motivation and recovery guidance.

2. Application Overview

App Name	Persist
Platform	Android
Framework	Flutter
Backend	Firebase Authentication and Cloud Firestore
AI Service	OpenRouter API using dart-define for local secret handling
Primary Purpose	Goal tracking, behavioral consistency, mood reflection, and insight generation
Current Deployment Scope	Release build prepared and tested locally; Play Store upload skipped with instructor approval

3. Lab Objectives Completed

- Prepared the mobile application for release build generation.
- Configured Firebase Authentication and Firestore for app functionality.
- Generated/verified Android build readiness using Flutter tools.
- Tested the app on a physical Android device.
- Prepared store-listing related material such as app icon, screenshots, and description content.
- Handled API key security by removing the OpenRouter API key from source code and using dart-define instead.
- Uploaded the project source code to GitHub after resolving secret-scanning issues.

4. Android Release Preparation

The Android release preparation process involved checking project configuration, ensuring the application builds correctly, confirming Firebase connectivity, and preparing the app for release build generation.

4.1 Package Name and App Identity

The application package name was verified from the Android project configuration. For a real Play Store release, the package name must be unique and should not be changed after publication. For this lab, the existing package name was considered acceptable for academic submission.

Screenshot 1: Android package/application ID

```

1  allprojects {
2      repositories {
3          google()
4          mavenCentral()
5      }
6  }
7
8  val newBuildDir: Directory =
9      rootProject.layout.buildDirectory
10         .dir("../..../build")
11         .get()
12  rootProject.layout.buildDirectory.value(newBuildDir)
13
14  subprojects {
15      val newSubprojectBuildDir: Directory = newBuildDir.dir
16      project.layout.buildDirectory.value(newSubprojectBuildDir)
17  }
18  subprojects {
19      project.evaluationDependsOn(":app")
20  }
21
22  tasks.register<Delete>("clean") {
23      delete(rootProject.layout.buildDirectory)
24  }
25

```

```

PS C:\Users\De11\Desktop\HH_Study\SUPERIOR_UNIVERSITY_DATA\Sem_6\MAD
\Lab\GIT_LABS\TIT601540_Project\Project\persist-flutter> flutter bui
ld apk --release --dart-define=OPENROUTER_API_KEY=sk-or-v1-28e10fb3d
800d3c6b7ab30dc1b7e76c899856d8e4e97cf2ae13ce9e54bbb6e95
Running Gradle task 'assembleRelease'...
 507.5s
✓ Built build\app\outputs\flutter-apk\app-release.apk (51.6MB)
PS C:\Users\De11\Desktop\HH_Study\SUPERIOR_UNIVERSITY_DATA\Sem_6\MAD
\Lab\GIT_LABS\TIT601540_Project\Project\persist-flutter>

```

android/app/build.gradle.kts shows the applicationId/package name.

4.2 App Icon and Branding

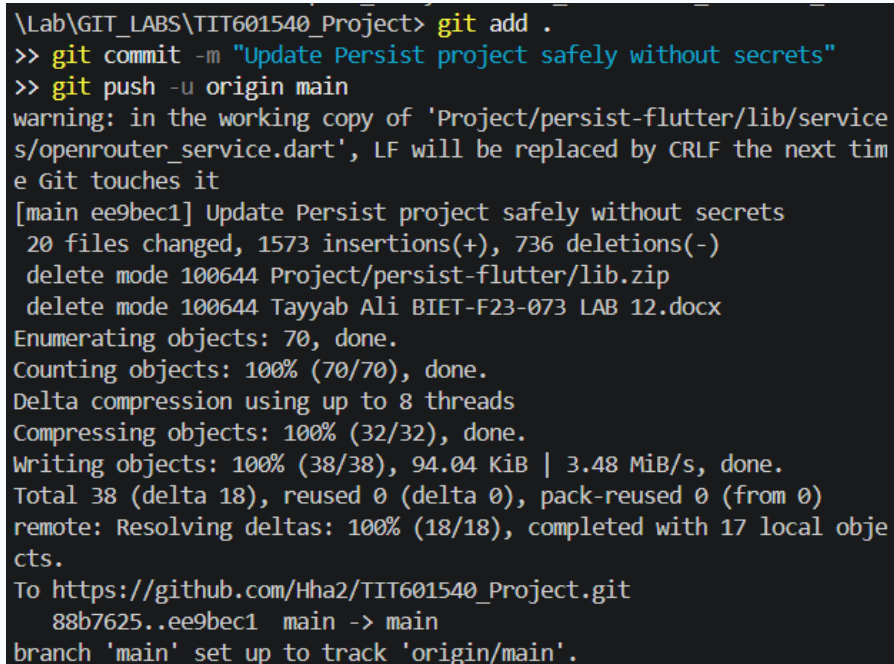
The app icon and loading screen were configured using the Persist logo. Android launcher icons were placed in the mipmap folders, and the app displays a branded loading screen during startup.

4.3 API Key Security

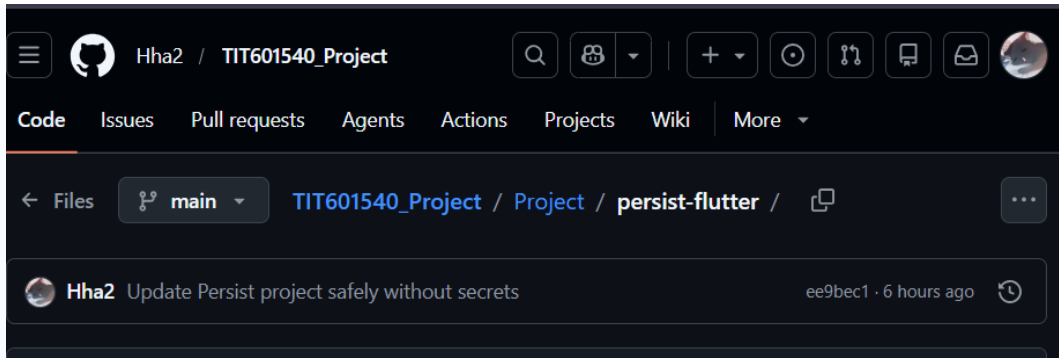
During development, GitHub push protection detected an OpenRouter API key in the source code. This was corrected by removing the hardcoded key and replacing it with a dart-define based configuration. This prevents secrets from being uploaded to the public GitHub repository.

```
const _apiKey = String.fromEnvironment('OPENROUTER_API_KEY');
```

Screenshot 2&3: GitHub push success / clean repository



```
\Lab\GIT_LABS\TIT601540_Project> git add .
>> git commit -m "Update Persist project safely without secrets"
>> git push -u origin main
warning: in the working copy of 'Project/persist-flutter/lib/service
s/openrouter_service.dart', LF will be replaced by CRLF the next tim
e Git touches it
[main ee9bec1] Update Persist project safely without secrets
 20 files changed, 1573 insertions(+), 736 deletions(-)
 delete mode 100644 Project/persist-flutter/lib.zip
 delete mode 100644 Tayyab Ali BIET-F23-073 LAB 12.docx
Enumerating objects: 70, done.
Counting objects: 100% (70/70), done.
Delta compression using up to 8 threads
Compressing objects: 100% (32/32), done.
Writing objects: 100% (38/38), 94.04 KiB | 3.48 MiB/s, done.
Total 38 (delta 18), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (18/18), completed with 17 local obje
cts.
To https://github.com/Hha2/TIT601540_Project.git
 88b7625..ee9bec1  main -> main
branch 'main' set up to track 'origin/main'.
```



Paste screenshot showing successful Git push after removing the hardcoded API key.

5. Release Build Generation

For Android release preparation, the recommended Play Store format is an Android App Bundle (.aab). However, for direct testing and submission evidence, a release APK can also be generated. Since the class was allowed to skip actual Google Play Console upload due to budget limitations, the release build and testing evidence are the main deliverables.

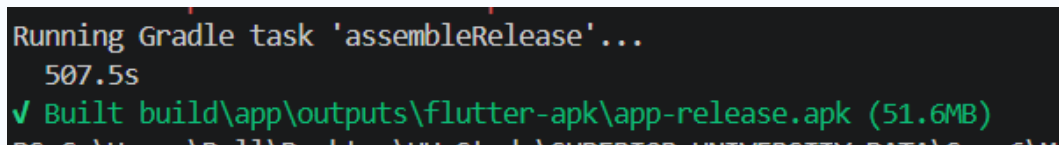
Command used to generate release APK:

```
flutter build apk --release --dart-define=OPENROUTER_API_KEY=YOUR_REAL_KEY
```

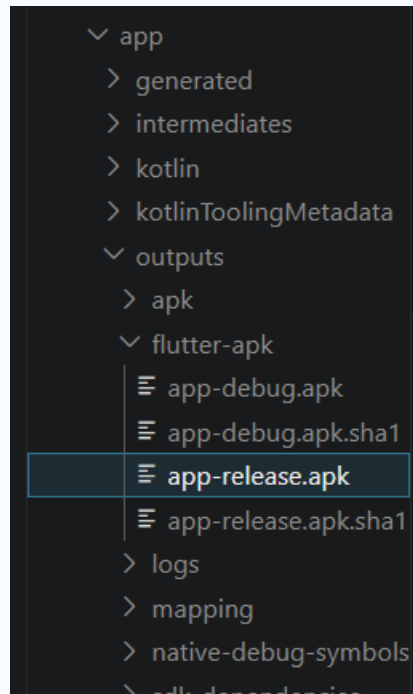
Optional command for Play Store App Bundle:

```
flutter build appbundle --release --dart-define=OPENROUTER_API_KEY=YOUR_REAL_KEY
```

Screenshot 4&5: Successful release APK/AAB build



Paste terminal screenshot showing successful flutter build apk or flutter build appbundle command.

Screenshot 6: Release build output folder

Paste screenshot of build/app/outputs/flutter-apk/app-release.apk or app-release.aab location.

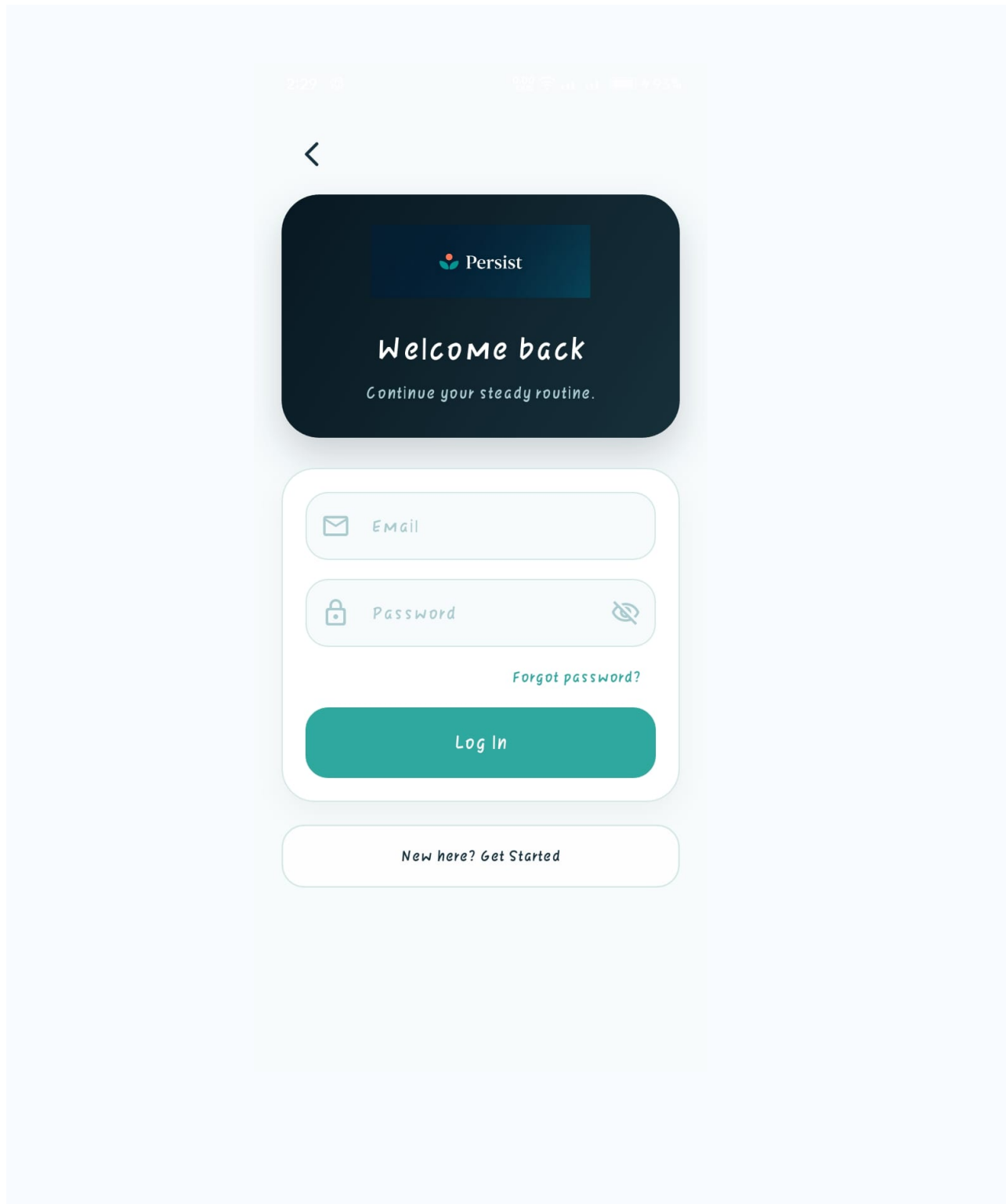
6. Physical Device Testing

The application was tested on a physical Android device. Testing on a real device is important because some features such as Firebase authentication, Firestore reads/writes, animations, app performance, and UI responsiveness behave more realistically on hardware than on an emulator.

Test Area	Expected Result	Status
Login / Authentication	User can sign in successfully using Firebase Authentication.	Completed / Tested
Home Screen	Home screen loads tasks, stability score, active goals, and behavior trend.	Completed / Tested

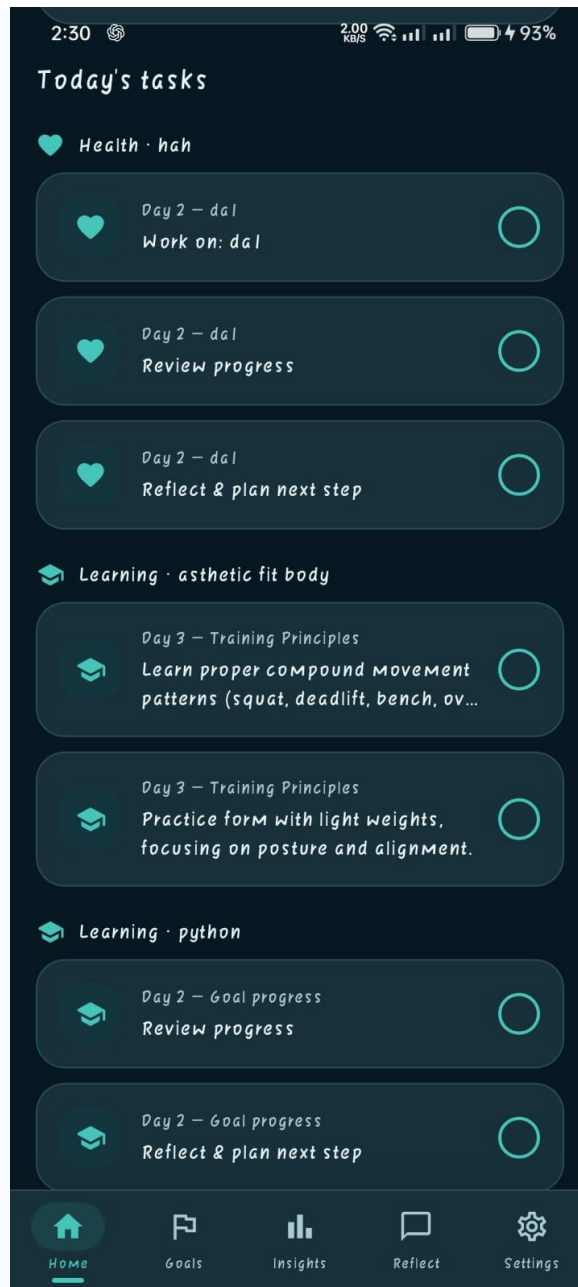
Manual Goal Creation	User can create structured goals with day subheadings and multiple tasks.	Completed / Tested
AI Goal Creation	AI can generate a day-wise goal plan using OpenRouter API.	Completed / Tested
Task Completion	Completed tasks disappear and day flow advances only at the correct time.	Completed / Tested
Mood Logging	Mood modal appears after task completion and logs mood feedback.	Completed / Tested
Insights Screen	Insights display stability, completion statistics, skip risk, and behavior patterns.	Completed / Tested
Reflection AI	AI reflection coach responds to user messages without duplicating welcome messages.	Completed / Tested
Settings Screen	Theme/settings and premium preview options are accessible.	Completed / Tested
GitHub Source Control	Project source code is committed and pushed after removing secrets.	Completed / Tested

Screenshot 7: Login screen



Screenshot 8: Home screen with today tasks





Screenshot 9: Goal creation screen

2:30 0.03 KB/S 93%

<

Create New Goal

Goal Name

e.g., Learn Flutter

Category

Learning Fitness Mindfulness

Career Health Creative

AI Duration

7 21 30 50 90

Difficulty

Easy Medium Hard

✦ AI Build for Me

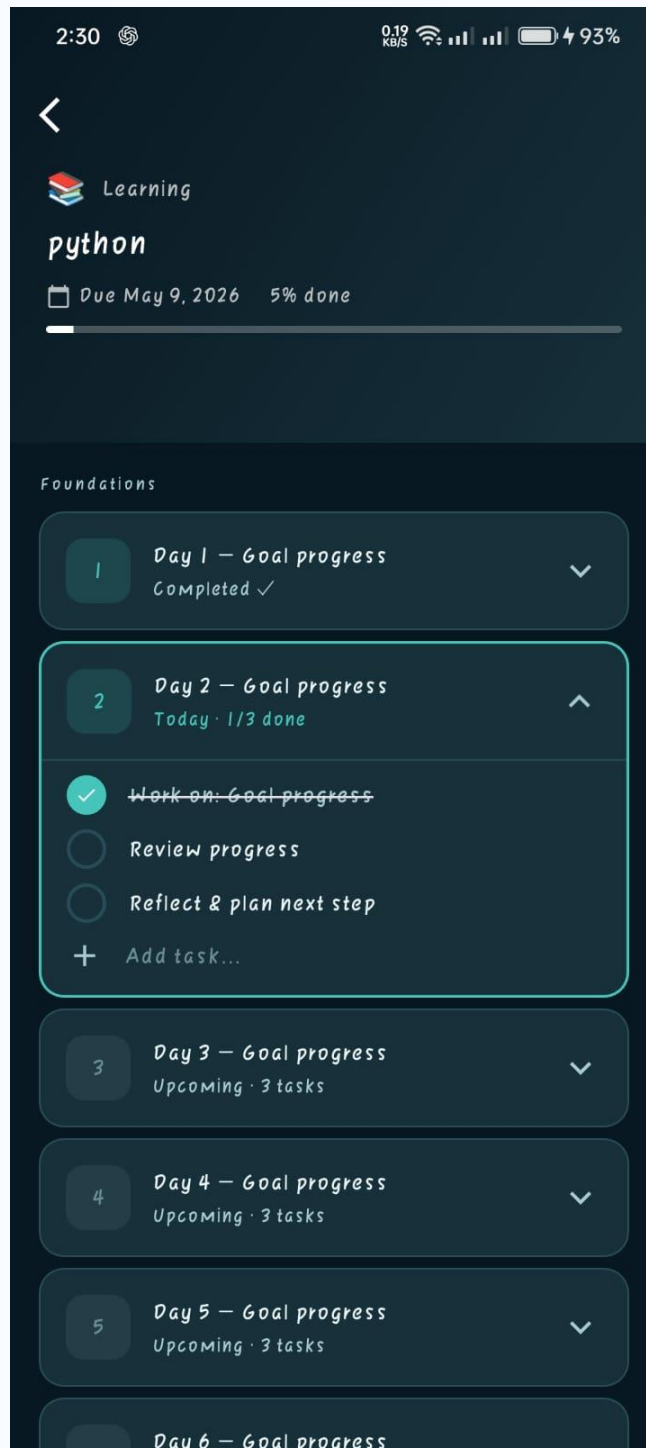
AI creates headings and tasks for every day.

Generate Plan

Manual Plan + Add Day

Each day needs a subheading and any number of tasks.

Screenshot 10: Goal detail screen



Screenshot 11: Insights screen



Screenshot 12: Reflection AI screen



7. Store Listing Preparation

Although actual Play Store submission was skipped with instructor approval, the required store listing material was reviewed and partially prepared. This includes the app name, short description, full description, app icon, screenshots, and privacy-policy requirement awareness.

App Name	Persist
Short Description	A goal tracking app that helps users stay consistent using tasks, mood logs, insights, and AI reflection.
Full Description	Persist helps users create day-wise goals, complete daily tasks, track mood, understand behavior patterns, estimate skip risk, and reflect with an AI coach to recover consistency.
Category	Productivity / Lifestyle
Pricing	Free
Privacy Policy	Required for real Play Store submission because Firebase and user data are used. For this lab, actual upload was skipped with instructor approval.

8. Google Play Console Submission Status

The original lab manual requires Google Play Console registration, Play Store listing creation, and internal testing track submission. However, the class discussed the budget issue with the instructor, and the instructor allowed students to skip the paid Google Play Console upload for now. Therefore, this submission focuses on release build preparation, real-device testing, source-code submission, and documentation of the publishing workflow.

Submission Status: Play Store upload skipped with instructor approval. Release preparation and testing completed.

9. Challenges Encountered and Solutions

Challenge	Solution
Firebase Firestore permission errors	Firestore rules were updated to allow authenticated users to access their own user document and subcollections.
AI chat not working initially	The placeholder OpenRouter API key and unavailable model were corrected. The API key was later moved out of source code using dart-define.
GitHub push blocked due to secret scanning	The hardcoded API key was removed from source code and duplicate folders/ZIPs containing the key were removed from Git history.
Task flow felt endless	The goal-day flow was updated so completing one day does not immediately show the next day. The next day unlocks later, while unfinished tasks remain visible.
UI overflow in goal detail screen	The header layout was adjusted to avoid Flutter RenderFlex overflow stripes.
Fake or weak insights	Insights were improved to use task completion, streak, overdue days, mood, and usage signals instead of purely decorative values.

10. Android Lab 13 Checklist

No.	Checklist Item	Status
1	Unique package name checked	Completed
2	App icon configured	Completed
3	Firebase connected	Completed
4	OpenRouter API key secured using dart-define	Completed

5	Release APK/AAB command prepared	Completed
6	App tested on physical Android device	Completed
7	Critical app screens tested	Completed
8	GitHub repository updated	Completed
9	Screenshots prepared for submission	To be pasted
10	Play Store upload	Skipped with instructor approval

11. Conclusion

Lab 13 provided practical understanding of how a mobile application is prepared for production deployment. Even though actual Google Play Store upload was skipped due to approved class budget limitations, the important technical parts of the deployment workflow were completed: release preparation, physical-device testing, app branding, build generation, security cleanup, and GitHub source control.

This lab also showed why deployment is not only about coding. Security, API keys, app identity, testing, release builds, and store requirements are equally important. The Persist app is now ready as a student project demonstration and can later be prepared for real Play Store publication by adding a privacy policy, creating a Google Play Developer account, uploading an AAB, and completing the Play Console listing.